

PARTS LIST

Semiconductors:

IC1	- ATmega8A MCU
IC2	- 7805 voltage regulator
IC3, IC4	- LM317 adjustable voltage regulator
D1, D2, D8, D12	- 1N5408 diode
D3-D7, D9-D11	- 1N4007 rectifier diode
T1, T2	- BC547 NPN transistor
LED1, LED2	- 5mm LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon), unless stated otherwise:

R1, R3-R4, R16, R22	- 1-kilo-ohm
R2, R9, R10, R17	- 2.2-kilo-ohm
R5-R7, R12, R13	- 4.7-kilo-ohm
R8, R18	- 1.2-ohm, 2-watt
R11, R19	- 47-kilo-ohm
R14, R20	- 470-ohm
R15, R21	- 150-ohm

Capacitors:

C1-C3, C5, C8, C9	- 0.1 μ F ceramic disk
C4	- 1000 μ F, 35V electrolytic
C6	- 100 μ F, 25V electrolytic
C7	- 10 μ F, 25V electrolytic

Miscellaneous:

CON1	- 6-pin connector
X1	- 230V AC primary to 9V-0-9V, 2A secondary centre-tap transformer
S1	- On/off switch
S2	- Tactile switch
RL1, RL2	- 9V, single-changeover relay
LCD1	- 16x2 LCD display
Batt.1, Batt.2	- 3.6V LiFePO4 for charging
	- Shorting jumper
	- 28-pin IC base
	- 2-pin terminal connector (two)
	- Heat-sink for IC2-IC4

EFY Note

The source code of this project is available for free download at source.efymag.com

'Full' message is displayed on LCD1. 'Absent' is displayed if the cell is not connected to the charging connector.

A good heat-sink should be provided separately for each LM317 to dissipate the heat generated in CC mode. **EFY**



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Construction and testing

An actual-size PCB layout for the automatic LiFePO4 Cell charger is shown in Fig. 2 and its component layout in Fig. 3. After assembling the circuit on PCB, connect the 9V-0-9V secondary output of the transformer across X1 marked on the PCB. Also connect the

primary of X1 to the AC mains via switch S1 (not shown in Fig. 3).

On switching on the power supply through power switch (S1), blinker LED1 starts blinking and LCD1 displays a welcome message. Then the program starts the loop of reading the cell voltage and sets the charging mode to CC/CV accordingly. If the cell is charged fully,

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